Deviance

Last week

- Exponential family
- GLM model fit

↓

IWLS

(Iterative weighted least squares)

- Variance of $\hat{\beta}$
- Estimation of $a(\phi)$ → Pearson's χ^2 statistic

Deviance & Residuals

- Deviance → model adequacy
- Deviance → model selection
- AIC
- Diagnostics

Deviance

(Remember → Binomial Poisson Deviance ≡ Scale Deviance.)

From here on we will w.l.o.g. adopt the convention that $a(\phi) = \phi$.

Definition: the scaled deviance for model A is

Notation simplification

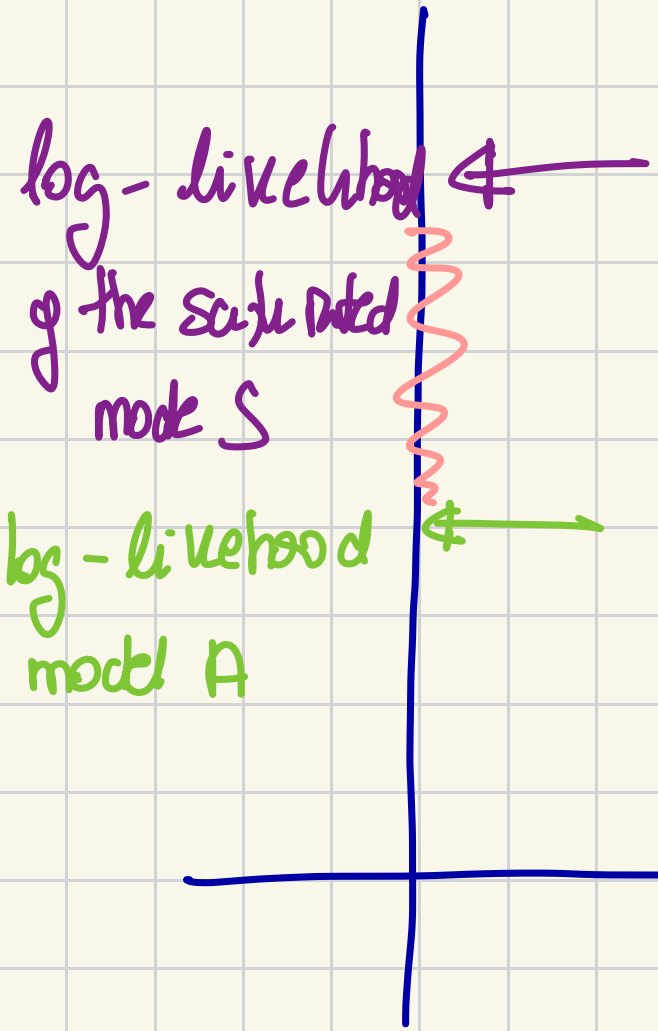
$$\frac{D^A}{\phi} = -2 \log \frac{\mathcal{L}(\hat{\beta}^A)}{\mathcal{L}(\mathrm{S})}$$

→ Maximum likelihood model A

→ Maximum likelihood saturated

where $\hat{\beta}^A$ is the MLE of β^A, the parameters in the model A, and $\mathcal{L}(\mathrm{S})$ is the maximum likelihood for the saturated model

The deviance is just D^A.



$$-2\log\left(\frac{\mathcal{L}(\hat{\beta}^A)}{\mathcal{L}_S}\right) \sim \chi^2_{n-p}$$

$$\frac{D^A}{\phi}$$

observations

parameters model A

DEVIANCE

$$\phi \, \frac{D^A}{\phi} = -2 \log \frac{\mathcal{L}(\hat{\beta}^A)}{\mathcal{L}(S)} \times \phi$$

- In the binomial case (Poisson): $\phi = 1$

  scale deviance = deviance !

Scale deviance for GLM: General case

- Model A: $Y \sim$ exponential family

$$E(y_i) = \mu_i = b'(\theta_i) = g^{-1}(x_i^t \beta)$$

MLE $\rightarrow \hat{\beta}^A \rightarrow \theta_i^A$ with ϕ known

Exponential family: $Y \sim$

$$f(y, \theta, \phi) = \exp\left[\frac{y\theta - b(\theta)}{a(\phi)} + c(y, \phi)\right]$$

$E(y) = b'(\theta)$

$Var(Y) = b''(\theta) a(\phi)$

- Saturated model: $S \rightarrow$ MLE $\hat{\theta}_i^S$, ϕ known

MLE $\longrightarrow$

Scale deviance :

$$\frac{D^A}{\phi} = -2 \log\left(\frac{\mathcal{L}(\hat{\beta}^A)}{\mathcal{L}(S)}\right)$$

$$= 2\left[\log \mathcal{L}(S) - \log \mathcal{L}(\hat{\beta}^A)\right]$$

$$= 2\left[\sum_{i=1}^{n} \frac{y_i \hat{\theta}_i^S - b(\hat{\theta}_i^S)}{\phi} + c(y_i, \phi) \right.$$

$$\left. - \sum_{i=1}^{n} \frac{y_i \hat{\theta}_i^A + b(\hat{\theta}_i^A)}{\phi} + c(y_i, \phi)\right]$$

Example: normal

The saturated normal model uses y_i to estimate μ_i.

The deviance can be written as $D = \sum_i d_i$ where

$$d_i = (y_i - \hat{\mu}_i)^2$$

where $\hat{\mu}_i$ is the fitted mean using the MLE.

It is equivalent to SSE (sum of squared errors) in linear models.

Example: normal

$Y \sim N(\mu, \sigma^2)$

$$\begin{aligned} f(y) &= \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{1}{2}\frac{(y-\mu)^2}{\sigma^2}} \\ &= \exp\left[\frac{y\mu - \mu^2/2}{\sigma^2} - \frac{1}{2}\left(\frac{y^2}{\sigma^2} + \log(2\pi\sigma^2)\right)\right] \\ &= \exp\left[\frac{y\theta - b(\theta)}{a(\phi)} + c(y,\phi)\right] \end{aligned}$$

where $\theta = \mu$, $\phi = \sigma^2$, and

$$\begin{aligned} b(\theta) &= \theta^2/2 \\ a(\phi) &= \phi \\ c(y,\phi) &= -\frac{1}{2}\left(\frac{y^2}{\phi} + \log(2\pi\phi)\right) \end{aligned}$$

For the Normal distribution:

$$\theta_i = \mu_i \quad , \quad b(\theta_i) = \frac{\theta_i^2}{2}$$

- Model A $\rightarrow$ $\hat{\theta}_i^A = \hat{\mu}_i^A = x_i^t \hat{\beta}_A$

  $\downarrow$ identity link function

- Saturated model S $\rightarrow$ $\hat{\theta}_i^S = \hat{\mu}_i^S = y_i$

\# parameters = \# observations

Scale deviance:

$$\frac{D^A}{\phi} = -2 \log\left(\frac{\mathcal{L}(\hat{\beta}^A)}{\mathcal{L}(S)}\right)$$

$$= 2\left[\log \mathcal{L}(S) - \log \mathcal{L}(\hat{\beta}^A)\right]$$

$$= 2\left[\sum_{i=1}^{n} \frac{y_i \hat{\theta}_i^S - b(\hat{\theta}_i^S)}{\phi} + c(y_i, \phi) - \sum_{i=1}^{n} \frac{y_i \hat{\theta}_i^A + b(\hat{\theta}_i^A)}{\phi} + c(y_i, \phi)\right] =$$

Saturated model

exponential fam. fy.

Model A

$$= \frac{2}{\phi}\left(\sum_{i=1}^{n} y_i(\hat{\theta}_i^S - \hat{\theta}_i^A) - (b(\hat{\theta}_i^S) - b(\hat{\theta}_i^A))\right)$$

$$= \frac{2}{\phi}\left(\sum_{i=1}^{n} y_i(y_i - \hat{\mu}_i^A) - \left(\frac{y_i^2}{2} - \frac{(\hat{\mu}_i^A)^2}{2}\right)\right)$$

$$= \frac{1}{\phi}\sum_{i=1}^{n} y_i^2 - 2y_i\hat{\mu}_i^A + (\hat{\mu}_i^A)^2$$

$$= \frac{1}{\phi}\sum_{i=1}^{n} (y_i - \hat{\mu}_i^A)^2$$

For the Normal distribution:

$$\theta_i = \mu_i \quad , \quad b(\theta_i) = \frac{\theta^2}{2}$$

Model A $\rightarrow$ $\hat{\theta}_i^A = \hat{\mu}_i^A = x_i^t \hat{\beta}_A$

(identity link function)

Saturated model $\rightarrow$ $\hat{\theta}_i^S = \hat{\mu}_i^S = y_i$

Deviance

$$\frac{D}{\phi} = \frac{1}{\phi} \sum_{i=1}^{n} \left(y_i - \hat{\mu}_i^A \right)$$

scaled deviance

$$\Rightarrow \quad \text{Deviance} \quad D = \sum_{i=1}^{n} \left(y_i - \hat{\mu}_i^A \right)$$

Example: Poisson → Exercise

The saturated Poisson model uses y_i to estimate $\mu_i = \lambda_i$.

The deviance can be written as $D = \sum_i d_i$ where

$$d_i = -2\left(y_i \log \frac{\hat{\mu}_i}{y_i} - (\hat{\mu}_i - y_i)\right)$$

where $\hat{\mu}_i$ is the fitted mean using the MLE.

Example: binomial → Exercise

The saturated binomial model uses y_i to estimate μ_i, that is y_i/m_i to estimate p_i.

The deviance can be written as $D = \sum_i d_i$ where

$$d_i = -2\left(y_i \log \frac{\hat{\mu}_i}{y_i} + (m_i - y_i) \log \frac{m_i - \hat{\mu}_i}{m_i - y_i}\right)$$

where $\hat{\mu}_i$ is the fitted mean using the MLE.

Scaled deviance for testing model adequacy

The binomial and Poisson models:

- $\phi = 1$ and the scaled deviance is just the deviance.
- When the individual responses are somewhat normal, if the model is adequate then the scaled deviance will be $\approx \chi^2_{n-p}$, where the saturated model has n parameters (equal to the number of observations) and the fitted model has p parameters.
- As a rule of thumb we need $m_i p_i$ and $m_i(1-p_i) \geq 5$ for the binomial model and $\lambda_i \geq 5$ for the Poisson model.

The normal, gamma or inverse gaussian models:

- we have to estimate ϕ using $X^2/(n-p)$. → Pearson's square statistic
- we can't use the scaled deviance to test model adequacy.

↳ Why ?

Scaled deviance for model selection (nested models)

The difference between two scaled deviances can be used for model selection between two nested models (i.e., testing the significance of the extra parameters).

The binomial and Poisson models: (see next page)

- For nested models, if the smaller model is correct then the difference between two scaled deviances is the log likelihood ratio and will be $\approx \chi^2_s$ where s is the difference in the number of parameters.
- Example : See "LR test using deviance" in Bliss.pdf

The normal, gamma or inverse gaussian models:

- we have to estimate ϕ using $X^2/(n-p)$.
- see the next slide..

(*) $LR = -2[\log \mathcal{L}(M_A) - \log \mathcal{L}(M_B)] \sim \chi^2_{\nu_B - \nu_A = s}$

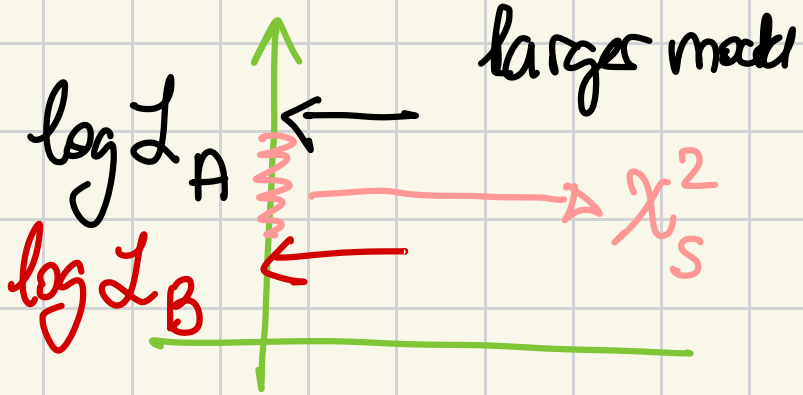
Case : $\phi = 1$ → Binomial and Poisson models

Nested models

Model 1 : $\eta_i = \beta_0 + \beta_1 x_i^1 + \beta_2 x_i^2$

Model 0 : $\eta_i = \beta_0$

1A is the larger model



Hypothesis test

H.T

$H_0 : \beta_1 = \beta_2 = 0$

$H_A : \exists i \ / \ \beta_i \neq 0 \quad i = 1, 2$

For nested models, if the smaller model is correct then the difference between two scaled deviances is the log likelihood ratio and will be $\approx \chi^2_s$ where s is the difference in the number of parameters.

(Bliss example)

Likelihood ratio

Model : $\eta_i = \beta_0 + \beta_1 x_i$

& Nested models

Model 0 : $\eta_i = \beta_0$ (no predictors)

Compare the deviance of the two models!

$$D^0 - D = -2\left[\log \mathcal{L}(\hat{\theta}^0) - \log \mathcal{L}(\hat{\theta})\right] \underset{H_0}{\sim} \chi^2_1$$

Anova

anova(model0, model, test = "Chi")

↳ analysis of deviance for glm models

Scaled deviance for model selection (nested models)

For the normal model (linear model), if model A is nested within model B, under the null hypothesis that model A is correct we have

$$\frac{(D^A - D^B)/s}{X^2/(n-p)} \sim F_{s,n-p}$$

where D^A and D^B denote deviances for models A and B, respectively, we have n observations, model A has $p - s$ parameters, and model B has p parameters.

smaller model

bigger model

- X^2 (Pearson's chi-squared) is calculated using model B.
- It is equivalent to the F-test in linear model.

For other models, this distributional result only holds approximately, but it can still be used for comparing two nested models. You will use it for comparing gamma models in the lab problem for the week 5.

$$\frac{(D^A - D^B)/s}{X^2/(n-p)} \sim F_{s,n-p}$$

$$\frac{D^A}{\phi} = -2 \log \frac{\mathcal{L}(\hat{\beta}^A)}{\mathcal{L}(S)} \times \phi$$

$$\frac{D_A - D_B}{\phi} = -2 \log \frac{\mathcal{L}(\hat{\beta}^A)}{\mathcal{L}(S)} - 2 \log \frac{\mathcal{L}(\hat{\beta}^B)}{\mathcal{L}(S)} / \phi$$

Intuition

Model A $<$ Model B

p−s parameters

p parameters

- $LR = \frac{D^A}{\phi} - \frac{D^B}{\phi} = \frac{D^A - D^B}{\phi} \sim \chi^2_s$ when ϕ is known
- If ϕ is unknown $\rightarrow$ $\hat{\phi} \rightarrow \frac{X^2}{n-p}$ computed using model B

$$\frac{\frac{D^A - D^B}{X^2}}{n-p} \rightarrow \frac{\frac{D_A - D_B}{s}}{\frac{X^2}{n-p}}$$

$\rightarrow \chi^2_s / s$

$\rightarrow \chi^2_{n-p}/n-p$

$\sim F_{s,n-p}$

I don't know the distribution

Test statistic

Scaled deviance for model selection (AIC)

The AIC is defined as

$$\text{AIC} = 2p - 2\log \mathcal{L}(\hat{\boldsymbol{\beta}})$$

where $\hat{\boldsymbol{\beta}}$ represent MLE of parameters in the model and p is the number of the parameters. Given a choice, we prefer the model with the smaller AIC.

If model B has s more parameters than model A (not necessarily nested within B), then

$$\begin{aligned} \text{AIC}^B - \text{AIC}^A &= 2s - 2\log \mathcal{L}(\hat{\boldsymbol{\beta}}^B) + 2\log \mathcal{L}(\hat{\boldsymbol{\beta}}^A) \\ &= 2s + \frac{D^B}{\phi} - \frac{D^A}{\phi}. \end{aligned}$$

difference in the number of parameters

→ Need to estimate ϕ.

Like the log likelihood ratio, the AIC needs an estimate of ϕ.

▶ AIC intuition ?

Wed

- Scale deviance
- Deviance

} GLM general case

→ Model adequacy → $\phi = 1$

- Model selection → Deviance (nested models)

$\phi = 1$ → log-likelihood ratio $\sim \chi^2_s$

$\phi \neq 1$ → estimate $\frac{D_A - D_B/s}{\chi^2/n-p} \sim F_{s,n-p}$

↘ AIC non-nested models

Today

- Continue with R example
- Diagnostics

Diagnostics

Diagnostics

- Check the adequacy of the model assumptions.

- Diagnostics → similar to those from linear models (with some adjustments!)
  - Understand/interpret different residual
  - Calculate the in R

Residuals ⟸ Response residuals
Pearson residuals
Deviance residuals

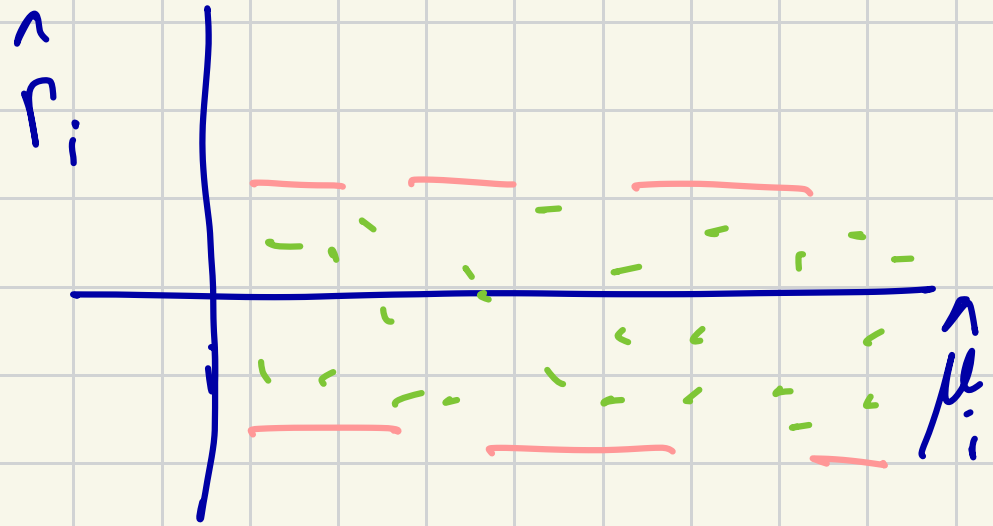
$$GLM \rightarrow \quad Y_i \;/\; \mathbb{E}(Y_i) = \mu_i = g^{-1}(x_i^t \beta)$$

$$r_i = y_i - \mu_i$$

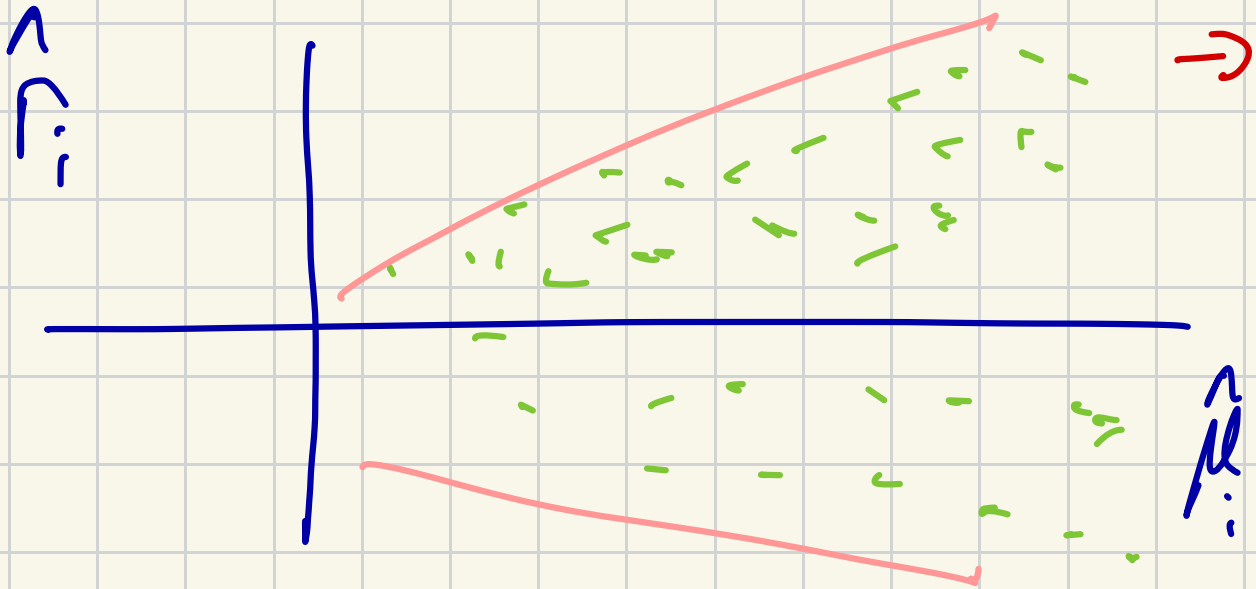
$$Var(r_i) = Var(y_i) = v(\mu_i)\, a(\phi)$$

not constant!

In the linear models $\longmapsto$ variance constant



→ constant variance



→ not constant variance).

Residuals

In general in GLM variance of y_i is not constant

Response residuals:

$$y_i - \hat{\mu}_i$$

Unless $v(\mu)$ is constant, as in the normal case, the response residuals are not homoskedastic and hence not very useful.

```
> residuals(glmfit, type="response")
```

$$\text{Var}\left(\frac{y_i - \mu_i}{\sqrt{v(\mu_i)}}\right) = a(\phi)$$

homoskedastic!

Pearson residuals:

$$r_P(i) = \frac{y_i - \hat{\mu}_i}{\sqrt{v(\hat{\mu}_i)}}$$

Pearson residuals are (approximately) homoskedastic, and $\sum_i r_P(i)^2 = X^2$.

$\hookrightarrow \hat{\phi}$

```
> residuals(glmfit, type="pearson")
```

Comparable to the standardized residuals in Linear Models

Residuals

$d_i = 2[l_i(S) - l_i(\hat{\beta}^{[1]})]$

Deviance residuals:

$$r_D(i) = \text{sign}(y_i - \hat{\mu}_i)\sqrt{d_i}$$

where the deviance is $D := \sum_i d_i = \sum_i r_D(i)^2$.

```
> residuals(glmfit)
```

→ default residuals

- As for linear models, patterns in the residuals indicate structure in the data that has not been captured by the model.
- We can plot the residuals against predictor variables, the responses, or the fitted means. Often a plot against $\hat{\eta}_i = \mathrm{x}_i^T \hat{\boldsymbol{\beta}}$ works well.
- With count data residual plots exhibit banding due to the discrete nature of the responses, and this can make it hard to see other patterns. In this case we can use a smoothed fit of the residuals to help spot trends/patterns.

Leverage in linear models

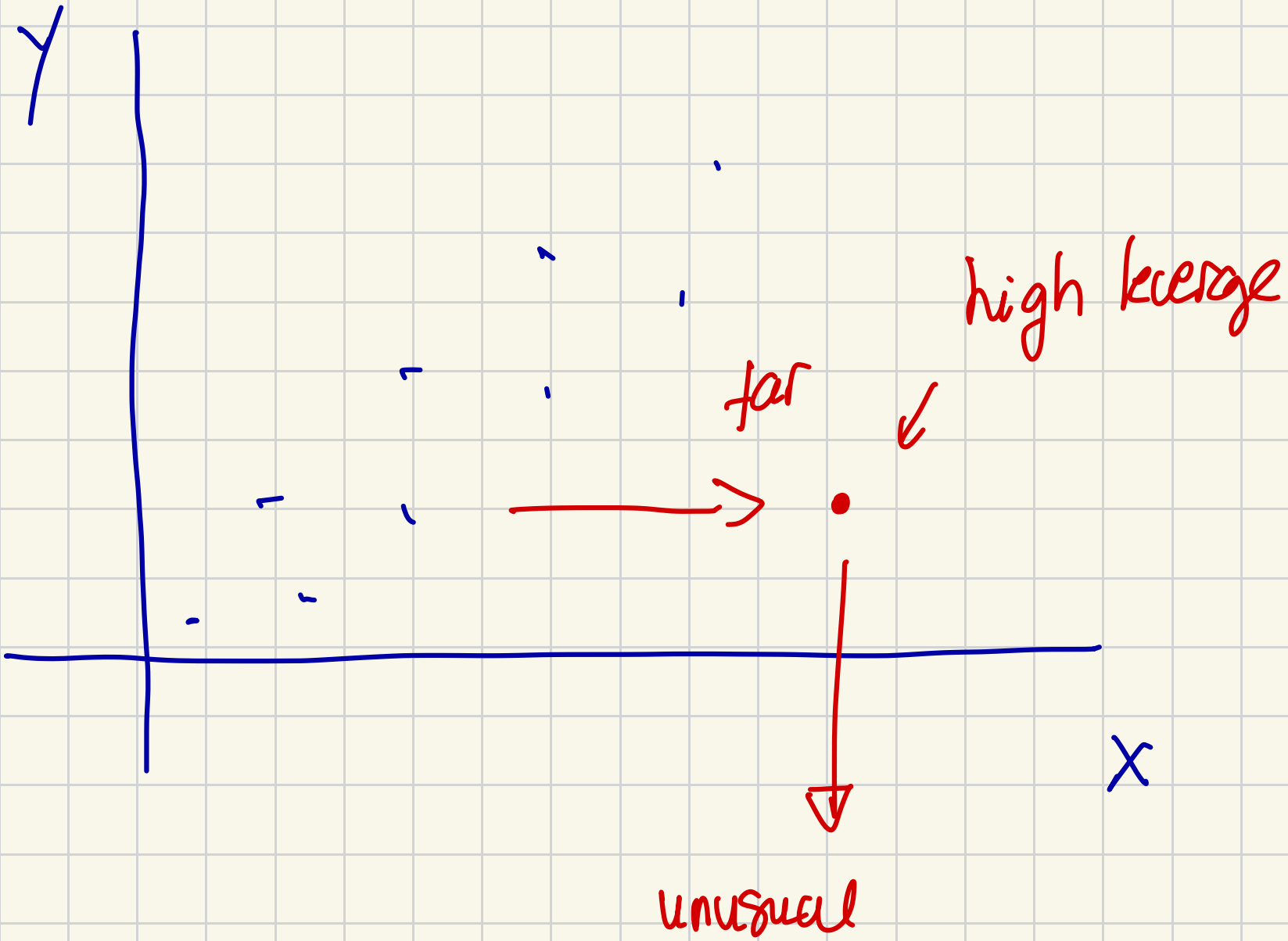
$$Y = X\beta + \varepsilon, \quad \varepsilon \sim N(0, \sigma^2 I)$$

$$\hat{Y} = X\hat{\beta} = \underbrace{X(X^tX)^{-1}X^t}_{H} y = Hy$$

$$H = \begin{bmatrix} h_{11} & & \\ & \ddots & \\ & & h_{nn} \end{bmatrix}$$

h_{ii} = measure the distance between $\bar{x}$ and X_i

Intuition: How far an observation's values are from the center of the x-values ($\bar{x}$) in the data


Y
high leverage
far
X
unusual

Leverage

The leverage measures the potential influence of a point on the fitted model. We borrow the definition of leverage from the theory of linear models, and use the hat matrix from the IWLS algorithm.

In the IWLS algorithm

$$\hat{Z} = X\hat{\beta} = X(X^T\hat{\Sigma}^{-1}X)^{-1}X^T\hat{\Sigma}^{-1}Z,$$

where $\hat{\Sigma}_{ii} = (g'(\hat{\mu}_i))^2 v(\hat{\mu}_i)\phi$ (assuming $a(\phi) = \phi$). Then, the hat matrix for $\hat{Z}$ is

$$H = X(X^T\hat{\Sigma}^{-1}X)^{-1}X^T\hat{\Sigma}^{-1}.$$

The leverage for the i-th observation is H_{ii}, the i-th diagonal element of H.

Note that H does not depend on ϕ (it cancels out).

A large leverage does not necessarily mean a point *has* influenced the fit.

```
> influence(glmfit)$hat
```

Thursday

Diagnostics

- Response residuals $r_i = y_i - \hat{\mu}_i$
- Pearson residuals $r_{p(i)} = \frac{y_i - \hat{\mu}_i}{\sqrt{v(\hat{\mu}_i)}}$

$$\sum_i r_{p(i)}^2 = X^2$$

- Deviance residuals $r_{D(i)} = \text{sign}(y_i - \hat{\mu}_i)\sqrt{d_i}$

$$D = \sum_i d_i = \sum_i r_{D(i)}^2$$

- Leverage → For observation i → H_{ii} (independent of ϕ)
  (potential influence of observation i – x-axis)

Today

Studentising → residuals in similar scale

Studentised residuals: $\text{Var}(z - \hat{z}) = (I - H)\Sigma$

Yesterday we showed $\left\langle \text{Var}(z_i - \hat{z}_i) = (1 - H_{ii}) \right.$ $\left. [g'(\mu_i)]^2 V(\mu_i)\phi \right.$

- Studentised Pearson residual $r_{sp(i)} = \dfrac{r_{p(i)}}{\sqrt{(1 - H_{ii})\hat{\phi}}} = \dfrac{y_i - \hat{\mu}_i}{\sqrt{(1 - H_{ii})V(\hat{\mu}_i)\hat{\phi}}}$ (leverage)
- Studentised deviance residual $r_{sD(i)} = \dfrac{r_D}{\sqrt{(1 - H_{ii})\hat{\phi}}}$ (leverage)

Intuition :

- Raw residual
- Pearson residual
- Deviance residuals

→ could be small due to the observation having a high leverage

→ Dividing by $\sqrt{1-H_{ii}}$ ensures their leverage is taking into account.

Intuition → How unusual y_i is, once we have adjusted for leverage, relative to model predictions

Interpretation

r_{sp}, r_{sD}

- close to 0 → observation is close to the fitted model ✓
- positive → y_i is larger than model prediction
- negative → y_i is smaller than model prediction

Studentised residuals

What is the variance of a residual from the model fit

$\hat{Z} = HZ \Rightarrow \mathrm{Var}(\hat{Z}) = H\,\mathrm{Var}(Z)\,H^t$

The variance of $\hat{Z}$ is

$$\begin{aligned} H\mathrm{Var}\,Z H^T &= [X(X^T\Sigma^{-1}X)^{-1}X^T\Sigma^{-1}]\Sigma[\Sigma^{-1}X(X^T\Sigma^{-1}X)^{-1}X^T] \\ &= X(X^T\Sigma^{-1}X)^{-1}X^T = H\Sigma. \end{aligned}$$

(*)

leverage

Whence $\mathrm{Var}\,(Z - \hat{Z}) = (I - H)\Sigma$.

Variance of residual depends on H

$\mathrm{Var}(Z - HZ) = \mathrm{Var}((I-H)Z) = (I-H)\,\Sigma\,(I-H)^t$

$= (I-H)\,\Sigma\,(I-H^t) = \Sigma - H\Sigma - \Sigma H^t + H\Sigma H^t = \Sigma - H\Sigma$

($H\Sigma$ = ΣH^t by (*))

Studentised residuals

From the above we have $\mathrm{Var}\,\frac{Z_i - \hat{Z}_i}{\sqrt{(1-H_{ii})g'(\mu_i)^2 v(\mu_i)\phi}} = 1$. Also

write it in terms of Y_i!

$$\begin{aligned} Z_i - \hat{Z}_i &= g(\mu_i) + (Y_i - \mu_i)g'(\mu_i) - [g(\mu_i) + (\hat{Y}_i - \mu_i)g'(\mu_i)] \\ &= (Y_i - \hat{Y}_i)g'(\mu_i) \end{aligned}$$

$\Rightarrow \mathrm{Var}(Z_i - \hat{Z}_i) = (1-H_{ii})g'(\mu_i)^2 v(\mu_i)\phi$

so, noting that $\hat{Y}_i = \hat{\mu}_i$,

$$\mathrm{Var}\,\frac{Y_i - \hat{\mu}_i}{\sqrt{(1-H_{ii})v(\hat{\mu}_i)\hat{\phi}}} \approx 1$$

The LHS is just $r_P(i)/\sqrt{(1-H_{ii})\hat{\phi}} =: r_{SP}(i)$, which we call the i-th **studentised Pearson residual**.

Studentised residuals

Studentised Pearson residual:

$$r_{SP}(i) = \frac{r_P(i)}{\sqrt{(1 - H_{ii})\hat{\phi}}}$$

By analogy we define **Studentised deviance residual:**

$$r_{SD}(i) = \frac{r_D(i)}{\sqrt{(1 - H_{ii})\hat{\phi}}}$$

Jack-knife residuals

A direct measure of the influence of a point is the **jack-knife residual**, which is the change in $\hat{\mu}_i$ when you remove y_i from the set of observations, then scaled to standardise the variance.

The jack-knife residual can be approximated by

$$\text{sign}(y_i - \hat{\mu}_i)\sqrt{(1 - H_{ii})r^2_{SD}(i) + H_{ii}r^2_{SP}(i)}.$$

```
> rstudent(glmfit)
```

Intuition: How much the fitted value for observation i changes when that observation is removed.

- Studentised residual: how unusual y_i is compared with model prediction
- Leverage: potential of an observation to affect the fit
- Jack-knife residual: what happens to the fit for observation i when we remove it.

Cook's distance

Another measure of the influence of the i-th observation is **Cook's distance**:

$$\frac{(\hat{\beta}^{(i)} - \hat{\beta})^T X^T \hat{\Sigma}^{-1} X (\hat{\beta}^{(i)} - \hat{\beta})}{p}$$

estimated without observation i

where $\hat{\beta}^{(i)}$ is the estimate of β obtained when y_i is omitted.

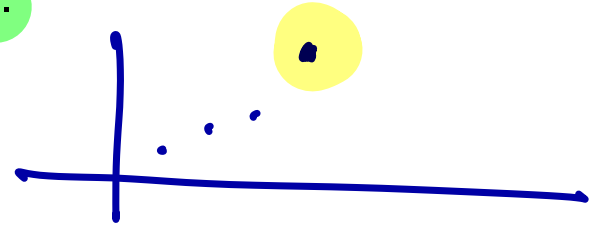
```
> cooks.distance(glmfit)
```

Intuition → How much the whole fitted model changes when observation i is removed.

Plotting leverage/jack-knife residuals/cook's distance

- When plotting it is helpful to consider them ordered by absolute size.
- If we plot the ordered absolute values against the percentage points of a half-normal distribution, then it is easier to see if the largest values are in keeping with the others. That is, plot the i-th ordered absolute residual against $\Phi^{-1}((n+i)/(2n+1))$, for $i = 1, \ldots, n$.
- If all is well we expect to see a smooth plot, while a jump or kink in the tail indicates a potential problem.
- Note that leverage/jack-knife residuals/cook's distance will not be normal, so don't expect a straight line.

As a reference scale for ordered absolute values



Checking linearity

- A non-linear link g (anything except the identity) makes it harder to check the assumption that $g(\mu_i) = \mathrm{x}_i^T \boldsymbol{\beta}$.
- The easiest thing to do is to plot $g(y_i)$ against $\{x_{ij}\}_{i=1}^n$ for each j and look for linear relationships.
- A more sophisticated approach is to plot $z_i = g(\mu_i) + (y_i - \mu_i)g'(\mu_i)$ against $\{x_{ij}\}_{i=1}^n$, where we use $\hat{\mu}_i$ for μ_i. From the IWLS algorithm we know that these plots should be linear.
- If there are non-linearities present then we can consider transforming $\{x_{ij}\}_{i=1}^n$. Transforming the responses y_i is often not a good idea for a glm, as this can break assumptions made about the distribution of Y_i.

Example

See Gala.pdf.

Learning goals

- Be able to define Generalised Linear Model (GLM).
- Be able to obtain the cannonical link function for GLM.
- (Challenging) Be able to explain ideas for the iterated weighted least squares (IWLS) algorithm.
- Be able to compute the variance of parameter estimates in GLM using IWLS algorithm.
- Understand (scaled) deviance
  - Be able to define (scaled) deviance.
  - Be able to compute (scaled) deviance.
  - Be able to use it to test model adequacy, perform model selection for nested models and non-nested models.
- Be able to perform diagnostics for GLM using R.
  - Understand what different 'diagnostics measurements' aim to measure.
  - Be able to obtain those diagnostic measurements from the *glm* output.
  - Be able to perform diagnostics for GLM using R.